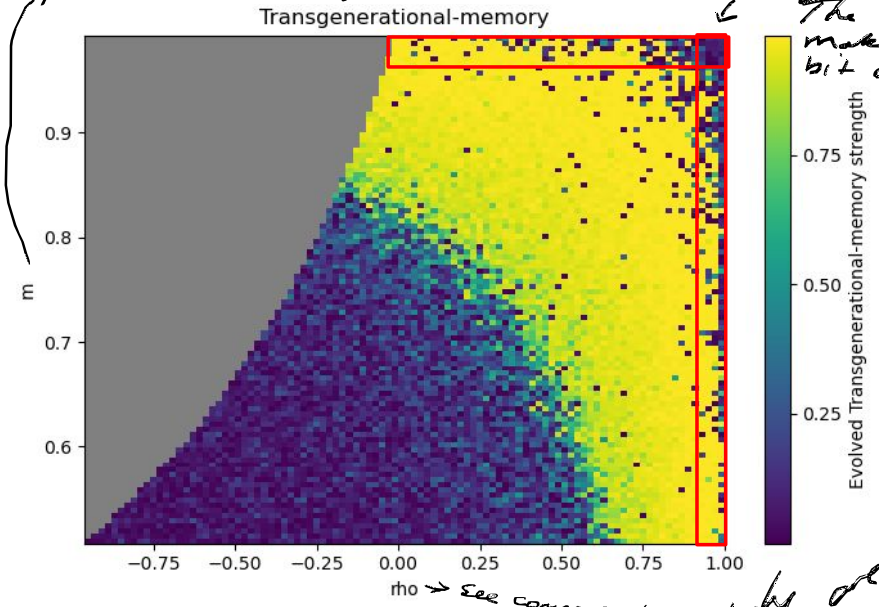
# Question 1: what does increasing dimension do to transgenerational memory

I want to show that the number of ‘worlds’ where memory evolves as well as the value of the evolved memory increases with increasing dimensions

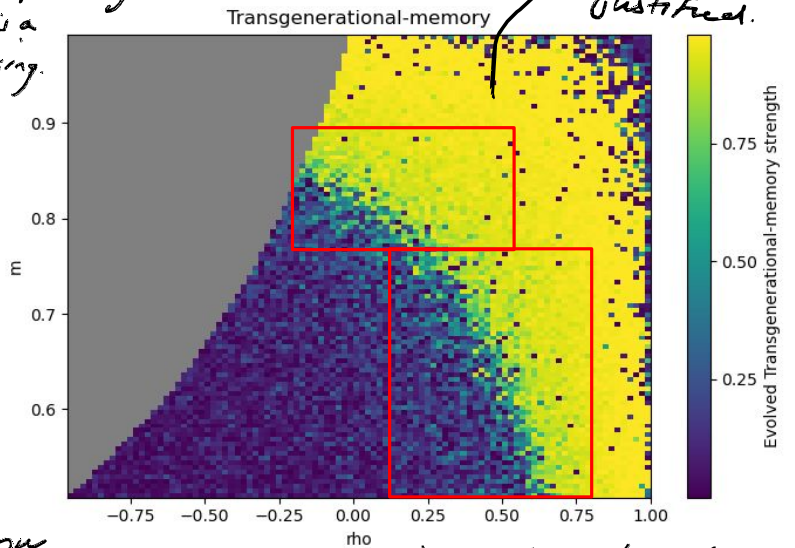
# Strategy 1: Count all the worlds in different portions of the heatmap that are > 0.75 (high) and <0.25 (low) memory

needs a more informative name for a table




it's just this square, right? The other rectangles make this a bit confusing.

This investigation is more easily justified.




See comment about m.

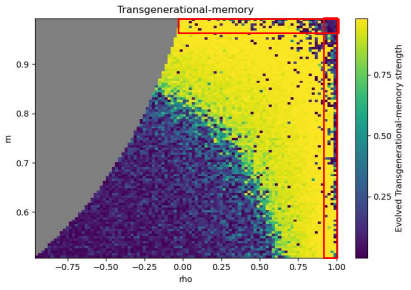
Zone 1: right corner

Why are you interested in this part of the parameter space? Should be biologically motivated.

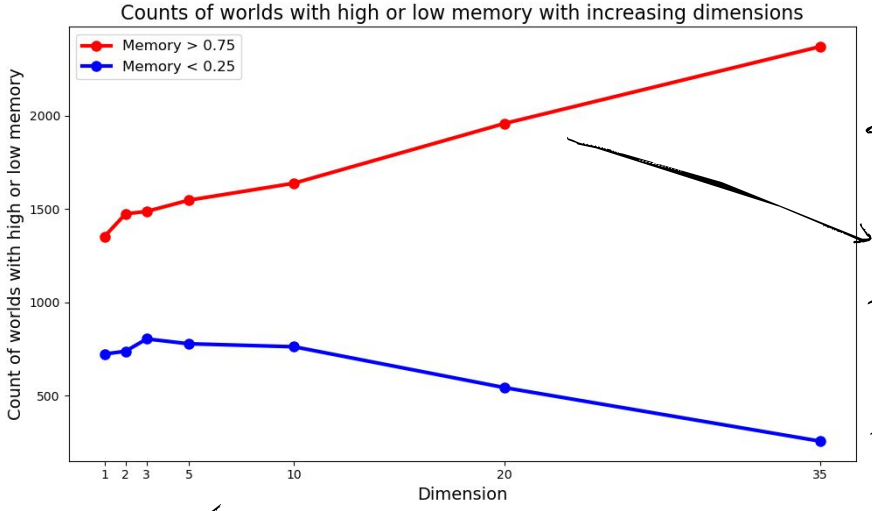
High time spent in one environment

Zone 2: Center strip

intermediate level of uncertainty...



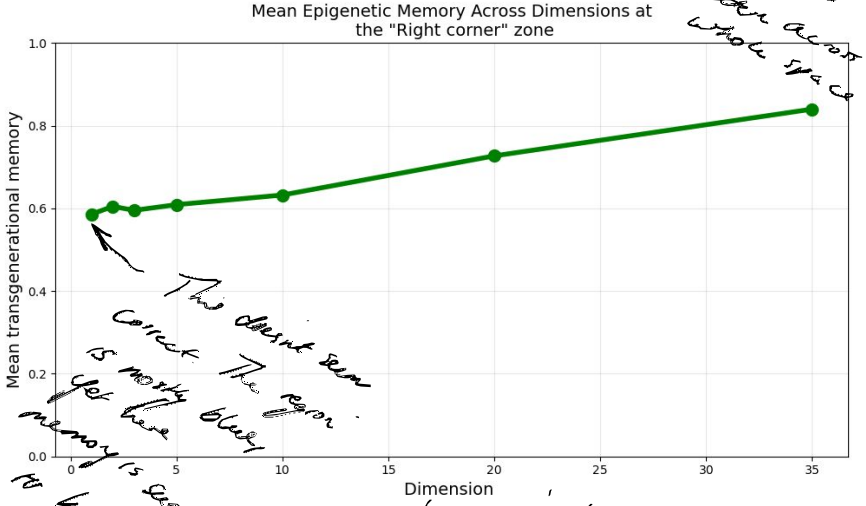

Zone 1: right corner




The hitting of all these plots are superlinear, all information is already in the axis title.

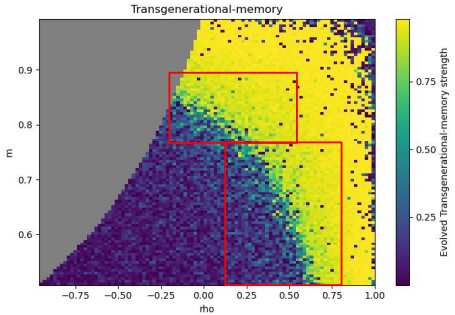
At higher dimensions, the increase is quite linear. Might not need so many dimensions if this is a general pattern across the whole space.

This could be on a log scale. Try and see, it may or may not be more effective than the current design.

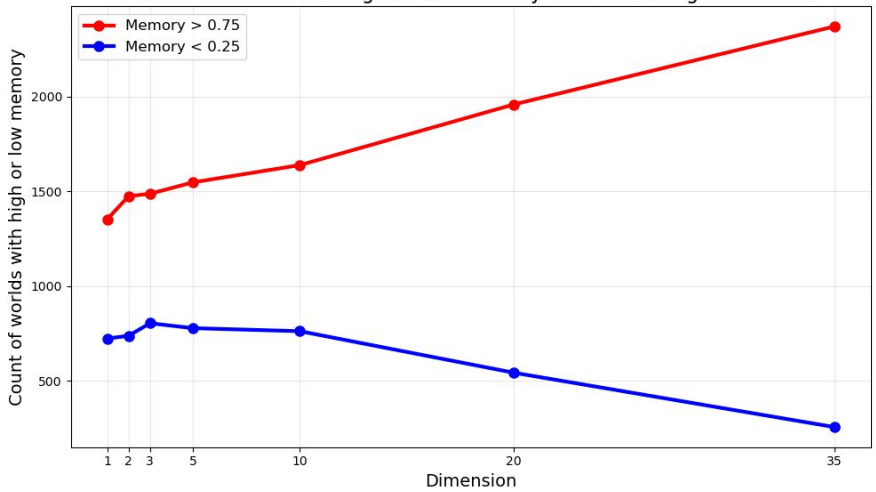



This doesn't seem correct. The region is mostly blue, yet here memory is suggested to be substantial.

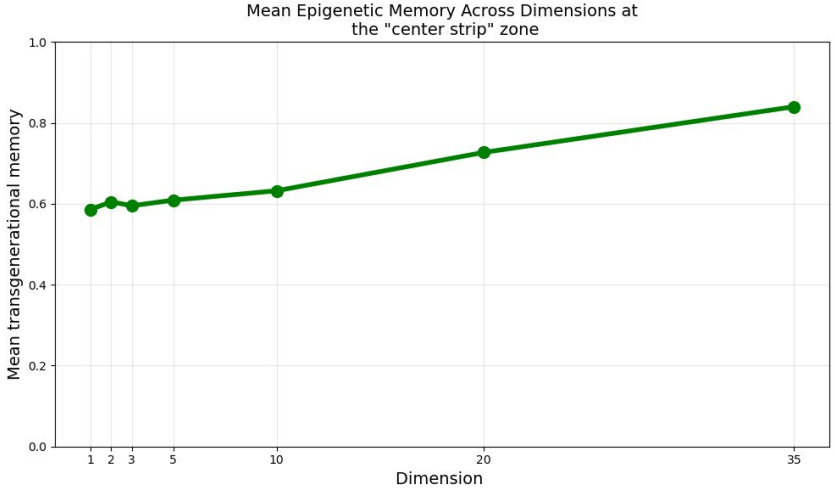
give better name.




Zone 2: Center strip



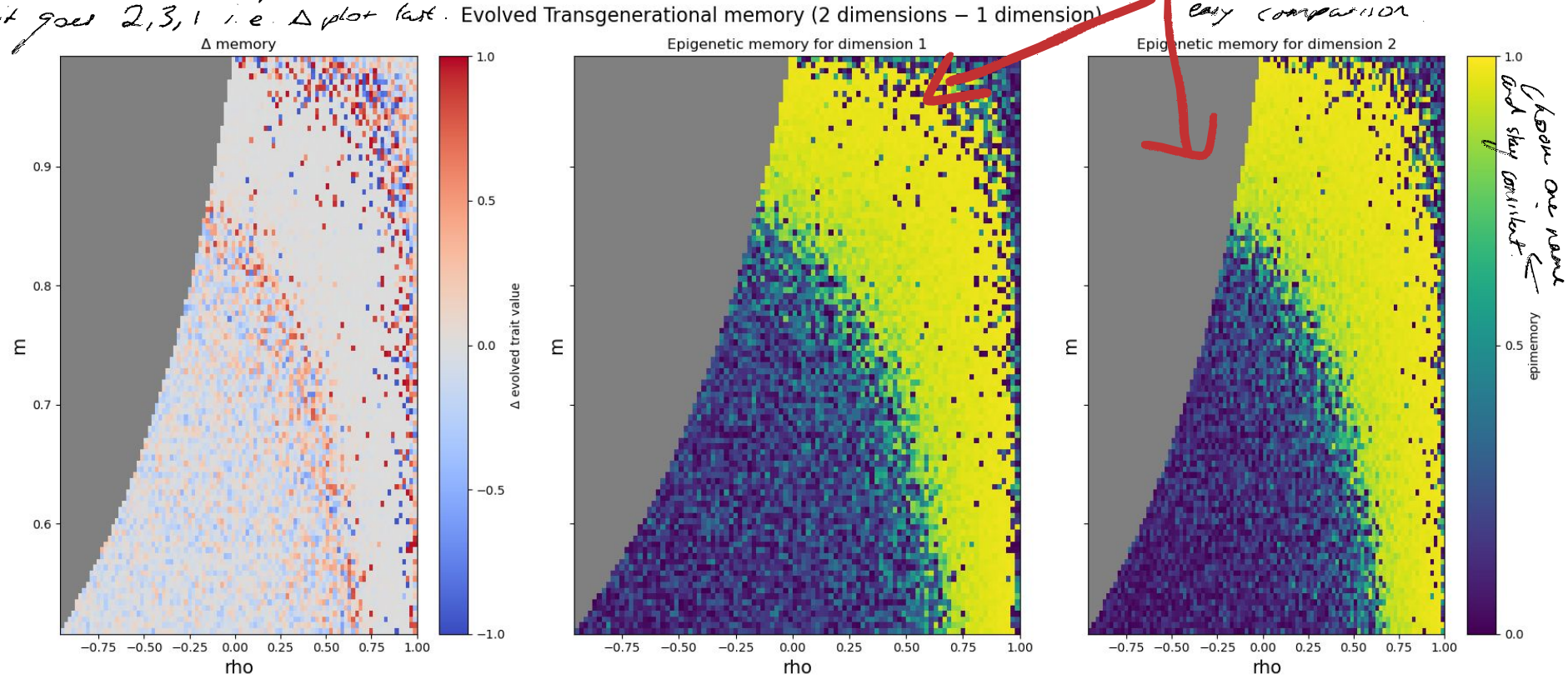

Thus are the same plots as the previous slide.



# Strategy 2: Compare the Memory(dim 'n') - Memory( dim 1) and see how this 'Δ' changes with increasing dimensions

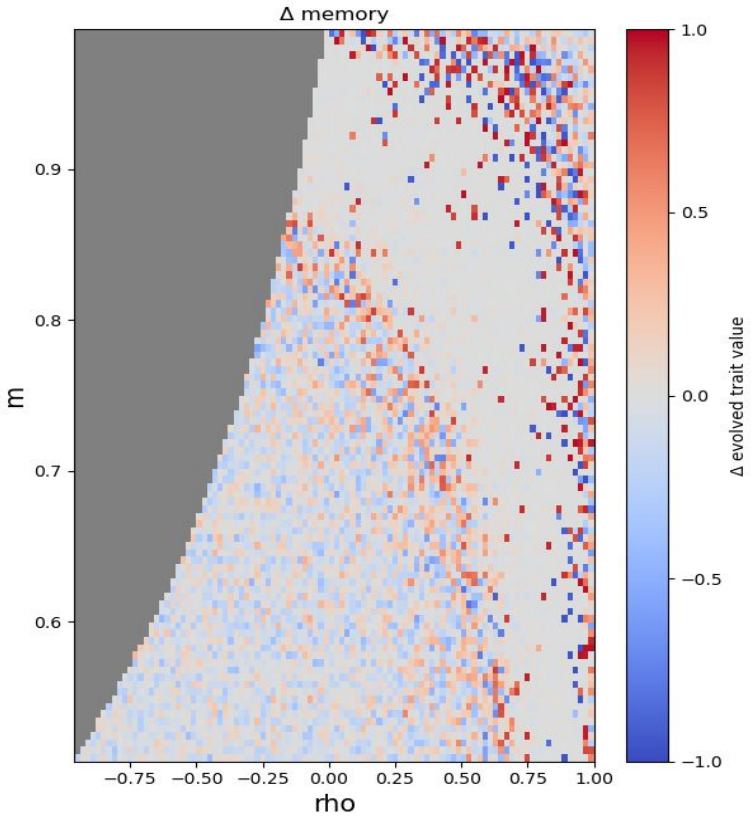
Swap order of plots so that it goes 2,3,1 i.e. Δ plot last.

These two plots need to be the same size for easy comparison




Here Δ= Higher dimension memory - 1D memory

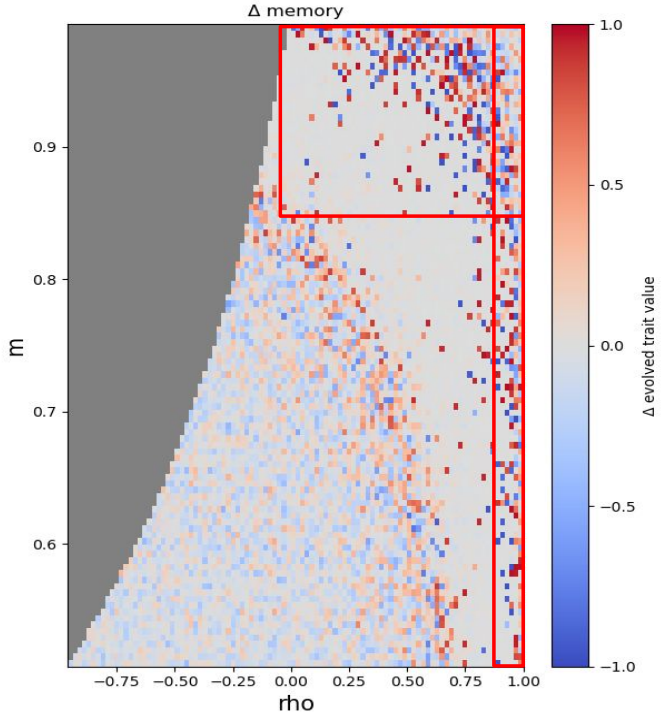
# Strategy 2: Use delta plots; Compare the Memory(dim ‘n’) - Memory( dim 1) and see how this ‘delta changes with increasing dimensions



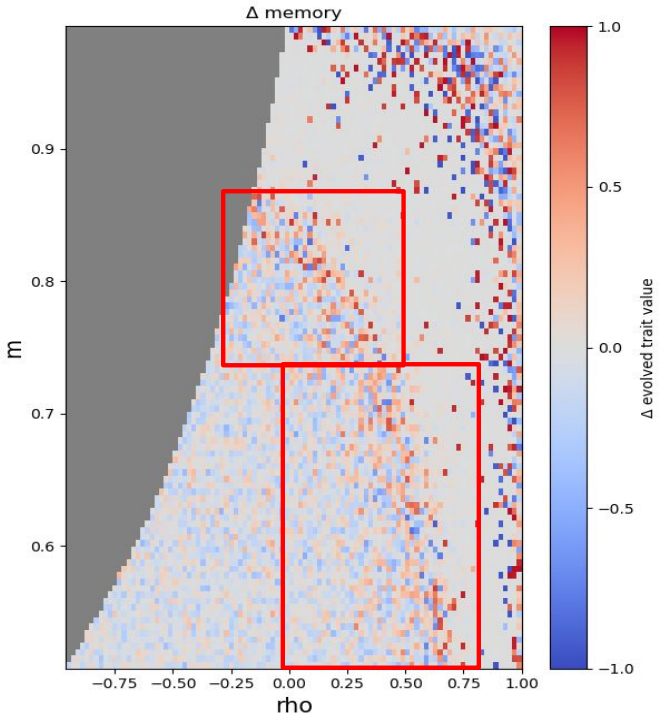

Positive Δ memory: digher dimension memory > 1 D memory

Negative Δ memory: higher dimension memory < 1 D memory

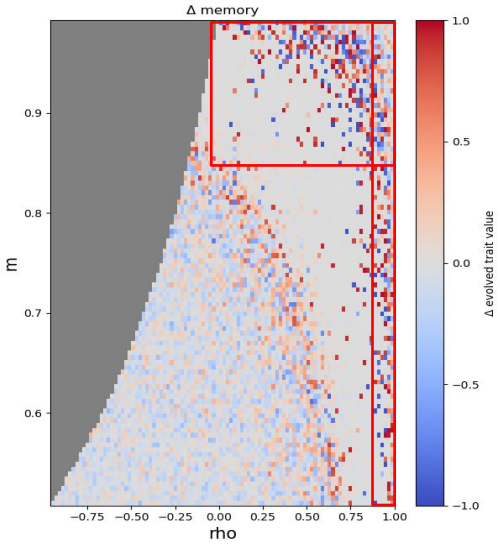
# Strategy 2: Compare the Memory(dim ‘n’) - Memory( dim 1) and see how this ‘delta changes with increasing dimensions

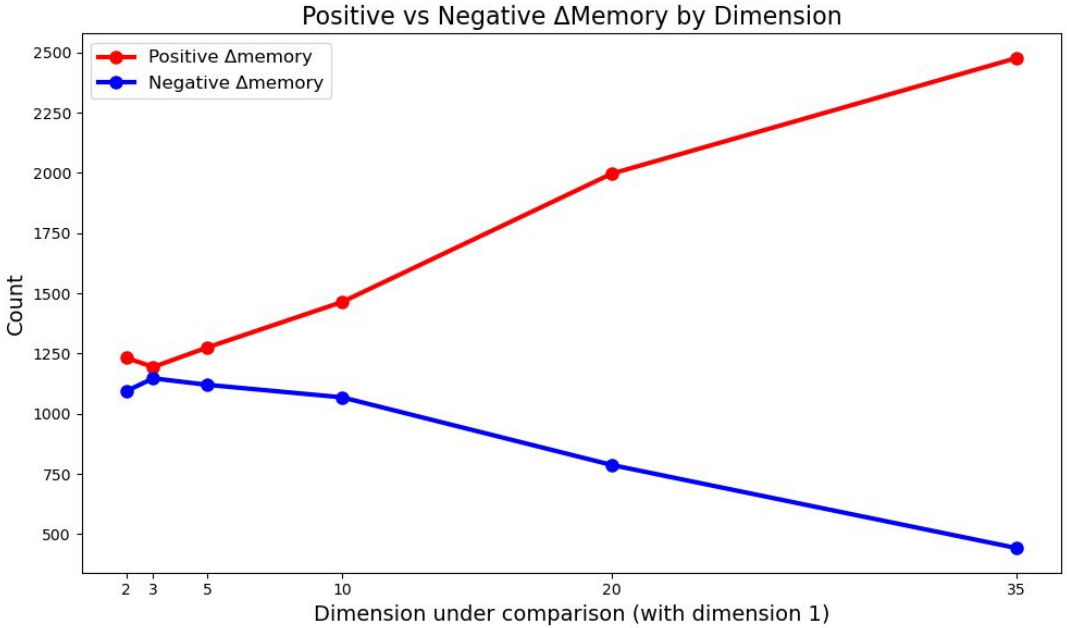



Zone 1: right corner

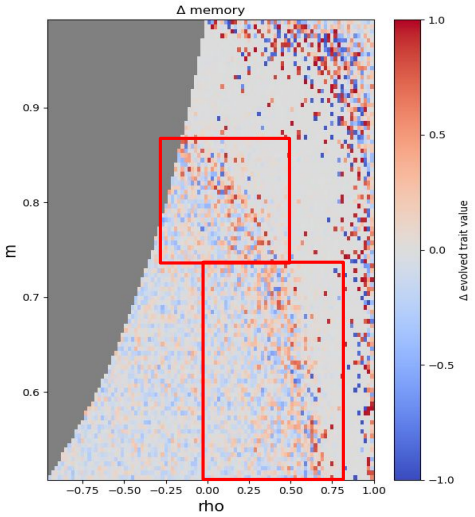



Zone 2: Center strip

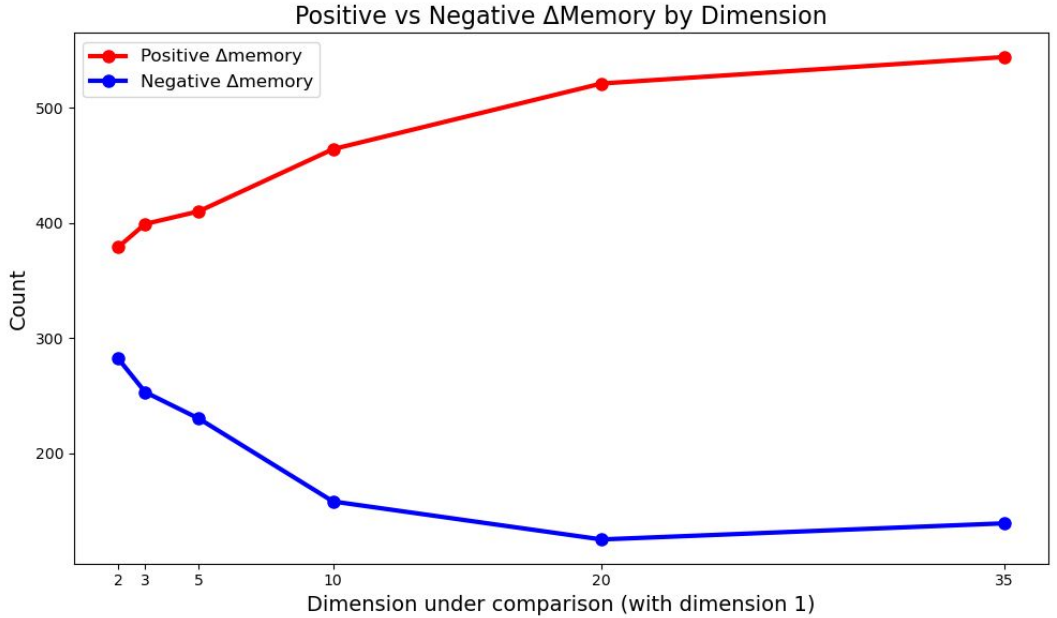



Here Δmemory = Higher dimension memory - 1D memory




Zone 2: Center strip




Here Δmemory = Higher dimension memory - 1D memory

# Question 2: How does the presence of memory change the fitness of a lineage across dimensions?

Try and investigate this by subtracting the (epi-memory fitness − only-genetic) fitness compared across dimensions

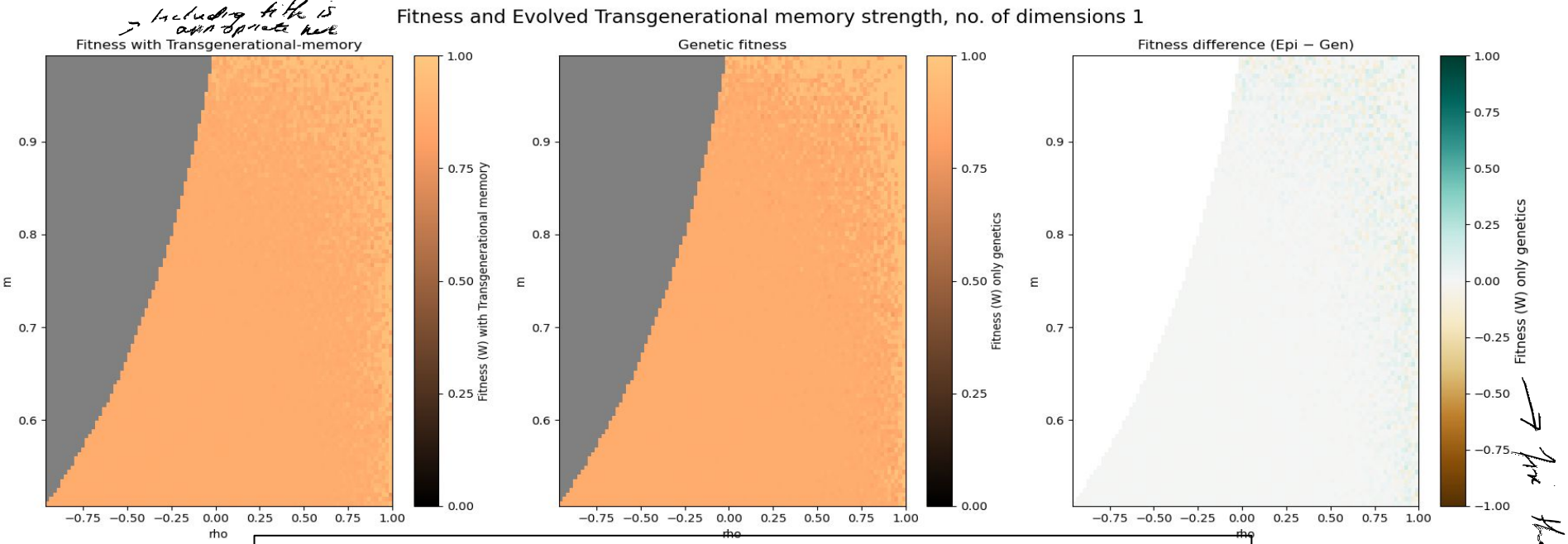
Here Δ Finess = Epi-fitness - Genetic fitness

↳ More on definitions. I would just call this a maternal effect eg. fitness with maternal effect. At the start have a slide where you define what you mean by maternal effect eg. epi...

# Does Δ Finess = Epi-fitness - Genetic fitness change with increasing dimensions?

Nice! Layout and plot sizes are very logical.

→ Including titles is appropriate here



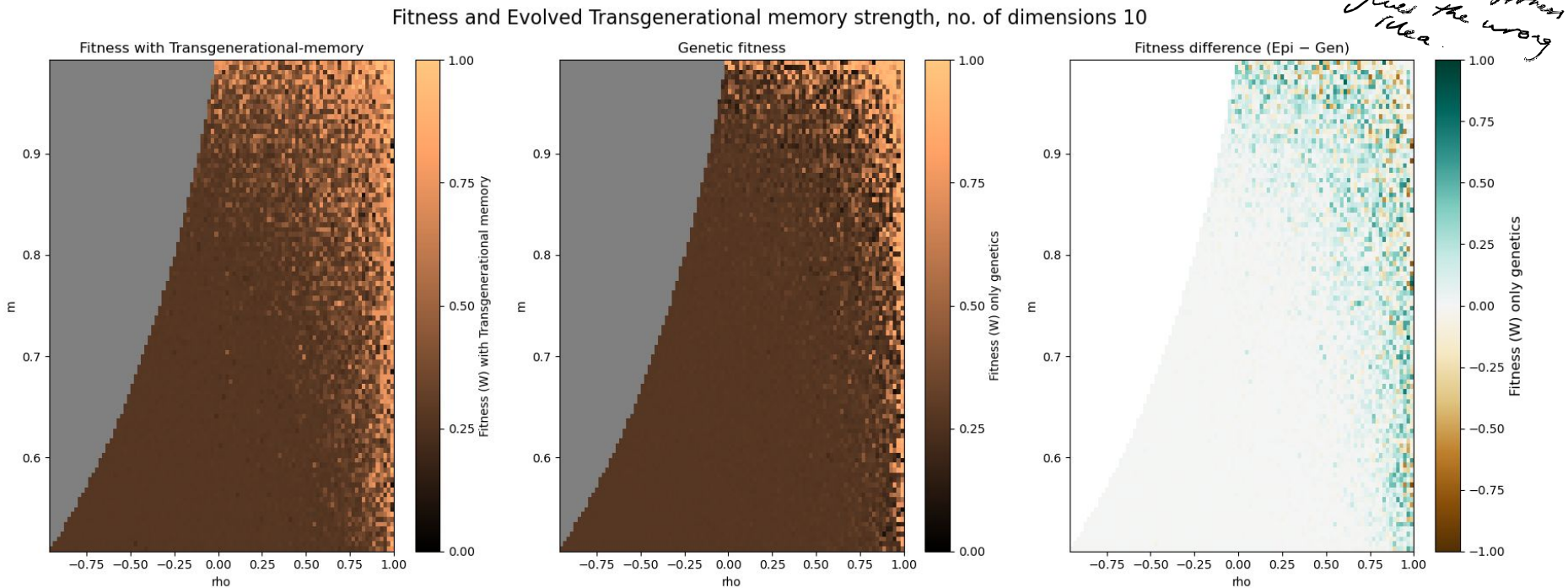

Dimension = 1

Here Δ Finess = Epi-fitness - Genetic fitness

→ Fitness following adaptation without maternal effects is more accurate

↳ Genetic fitness gives the wrong idea

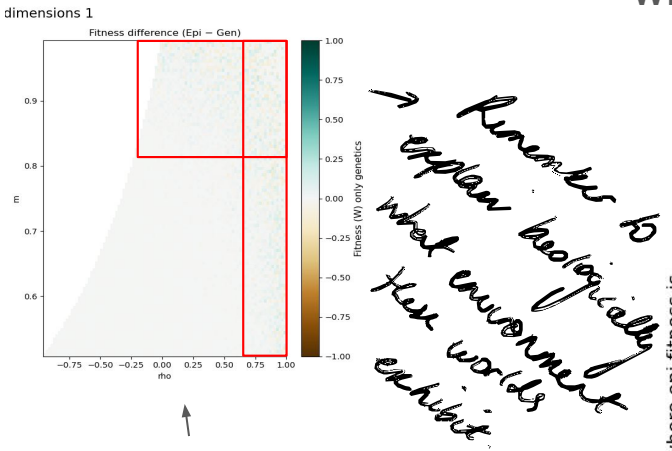
# Does Δ Finess = Epi-fitness - Genetic fitness increase or decrease with increasing dimensions




Dimension = 10

Here Δ Finess = Epi-fitness - Genetic fitness

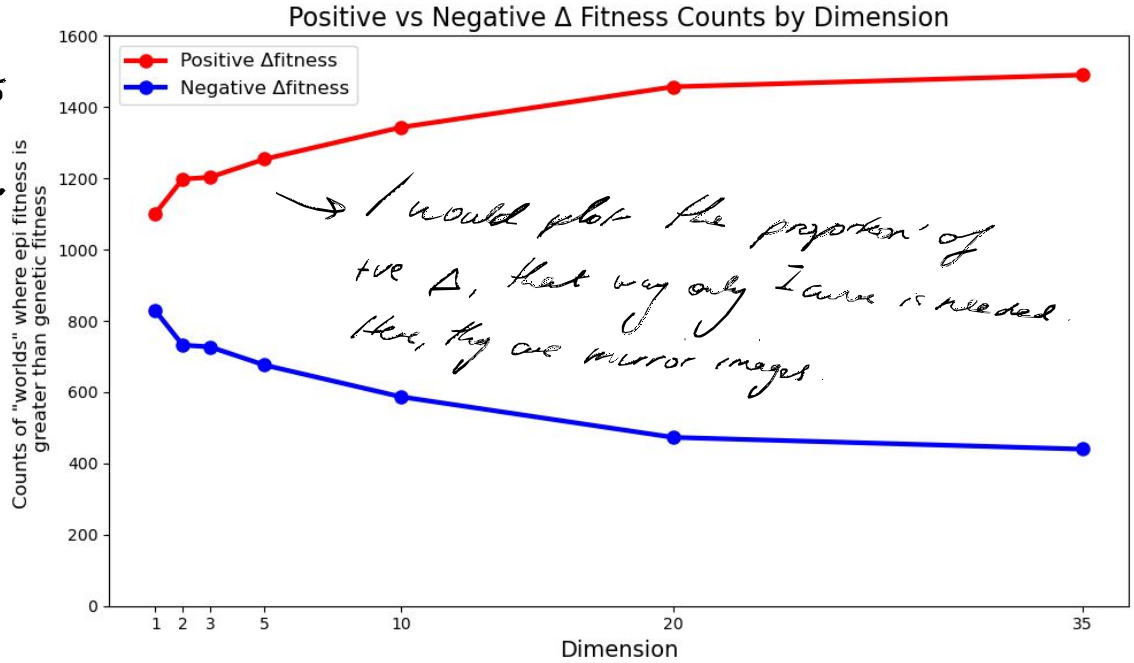
# Does (Δ Finess = Epi-fitness - Genetic fitness) of the **right corner** increase or decrease with increasing dimensions




→ Remember to explain biologically what environment these worlds exhibit

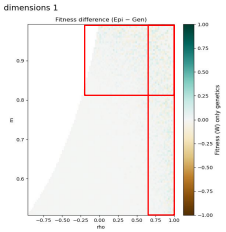
**Right corner worlds:** whats happening here? - Does the number of worlds with positive Δ Fitness (= Epi-fitness - Genetic) change with increasing fitness?

→ This is a bit redundant with the title. Pick 1 to reduce clutter.

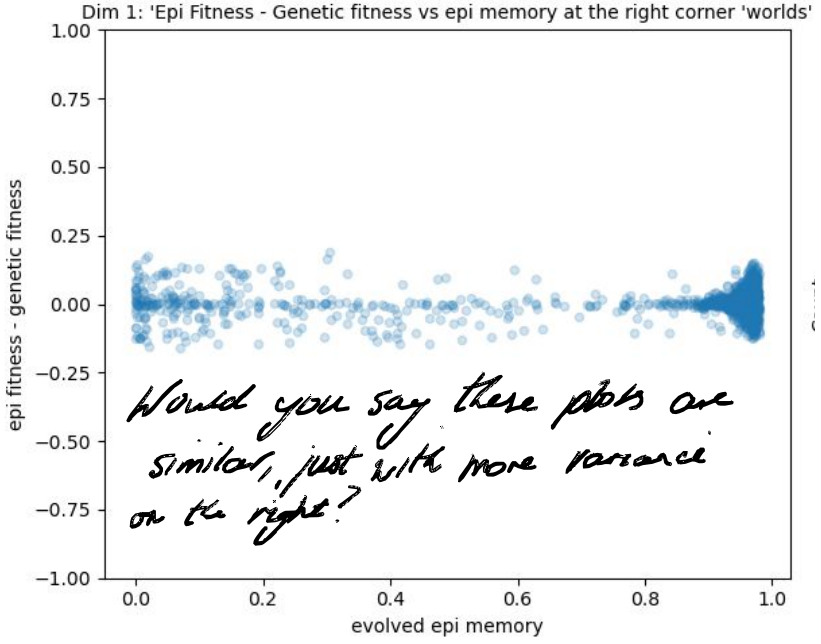



→ I would plot the proportion of +ve Δ, that way only 1 curve is needed. Here, they are mirror images.

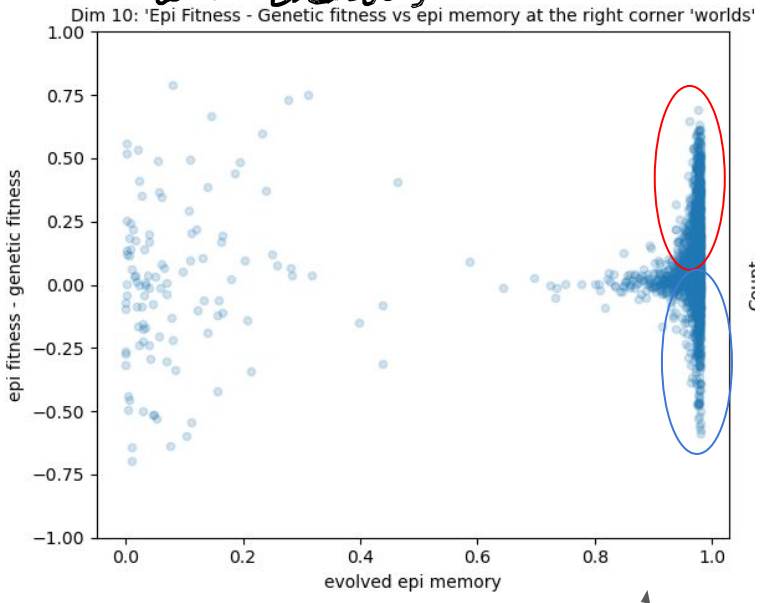
# Then I examine how the Δ Finess = Epi-fitness - Genetic correlates with the value of evolved memory at the ‘right corner’




bar plots here, rather than dot plots, might get the message across better. That way we can see that the proportions are changing.



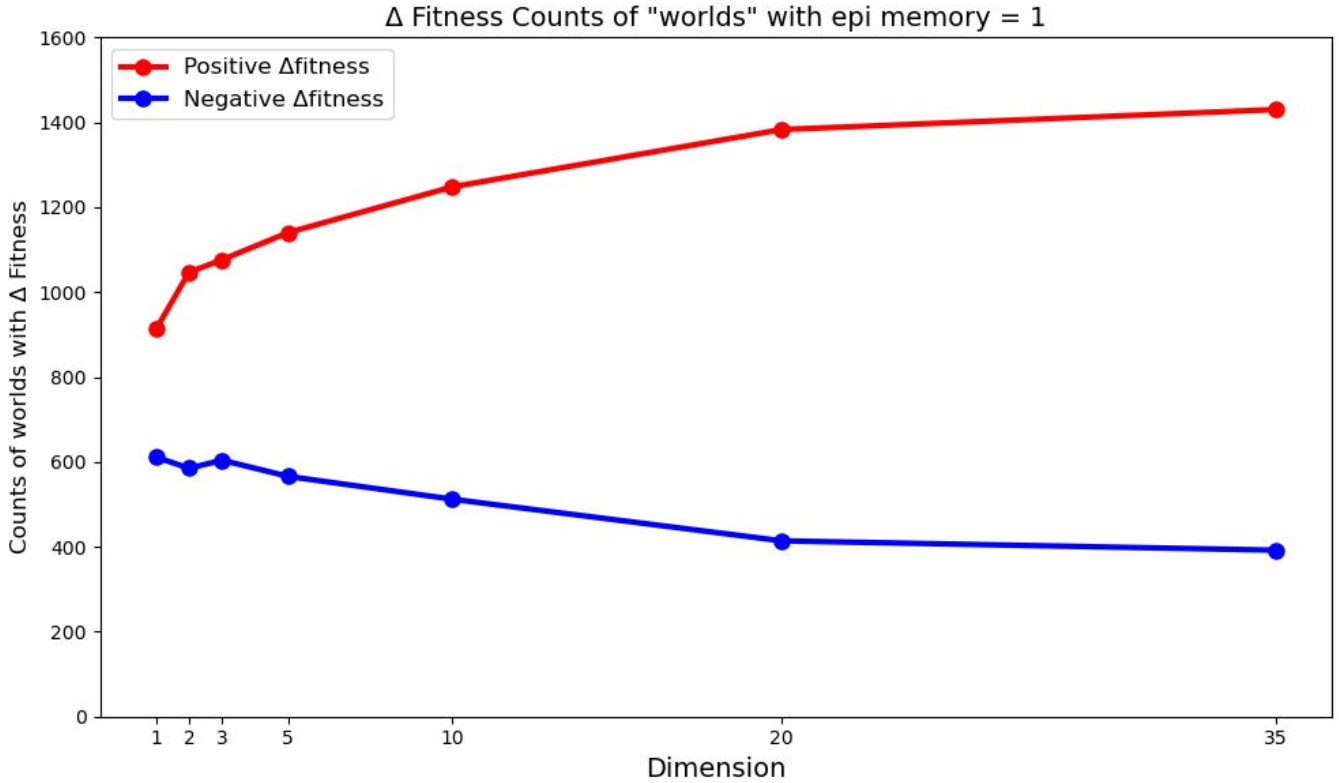

Would you say these plots are similar, just with more variance on the right?




I don't think this is much of a correlation. More, there are lots of cases where memory evolves to be high, and many AW effects are possible.

Both positive and negative Δ Fitness correlates with high (~ 1) epi memory in this region

# Subsetting ‘worlds’ in the right corner with Epi memory = 1




Prop. worlds with +ve Δ.

Another case where plotting the proportion is more efficient.

But when added up, on average high memory tends to have positive Δ Fitness than negative Δ Fitness, and this effect increases with increasing dimensions

# Results summary

Good for a real talk, but probably not required for this more informal setting.

1. Higher dimensional phenotypes expands the number of environmental parameter combinations ("worlds") that evolve positive transgenerational memory as well as the value of evolved memory in these worlds.
2. In the high-ρ, high-m parameter region, memory evolution outperforms genetics-alone strategies across more worlds. This advantage grows with phenotypic dimensionality, as higher dimensions expand the number of worlds where memory yields higher fitness.
3. High memory lineages (memory ~ 1) in the high-ρ, high-m region on average tend to have higher fitness than their genetic counterparts; and this effect amplifies with increasing dimensions
    a. High memory lineages in the rest of the parameter space show no increase in fitness in the presence of memory

1. Memory + dimensionality = adaptive in stable environments
2. Memory+ diemnsionality = still evolves in unstable environments:  but why? – maybe explanations involving positive frequency dependence requires invoking